FIELD OPERATIONAL NARRATIVE

UPTOWARE FLEET ERP
LIVE DEMONSTRATION SCENARIOS

A Complete 24-Hour Operational & Financial Lifecycle of Heavy Tipper KA-19-MG-8802 at Kudremukh North Pit

KALINGA MINING & LOGISTICS | FIELD-VALIDATED DATA | AUDIT-TRAIL VERIFIED

FICTIONAL ENTERPRISE ANCHOR
Kalinga Mining & Logistics Ltd.

PRIMARY OPERATIONAL SITE
Kudremukh North Pit (Iron Ore Crushing Circuit)

HAULAGE ASSET UNDER TEST
BharatBenz 3528CM 8x4 Tipper • Reg: KA-19-MG-8802

CERTIFIED DRIVER OPERATOR
Rajesh Kumar • Heavy Commercial Licence: DL-042018009124

1. Scenario Master Data Architecture

To demonstrate software convincingly to mining clients, you must tell a cohesive operational story where every number connects. This scenario tracks **Tipper KA-19-MG-8802** from morning shift roster to evening financial profitability.

ASSET & PIT PARAMETERS

Vehicle: BharatBenz 3528CM Tipper (KA-19-MG-8802)
Payload Rating: 35.00 Metric Tons Maximum
Unladen Tare Weight: 18.00 Metric Tons
Extraction Pit: Kudremukh North Pit (Bench 4)
Mineral Hauled: High-Grade Iron Ore Lump (62% Fe)
Destination: Railway Siding #2 Stockpile (18 km Haul)

COMMERCIAL & BILLING PARAMETERS

Contract Client: Tata Steel Mining Operations
Agreed Freight Rate: ₹450.00 per Metric Ton delivered
Diesel Bunkering Rate: ₹92.50 per Litre (Internal Pit Bowser)
Expected Fuel Mileage: 2.0 to 2.4 km per Litre
Driver Shift Bhatta: ₹450.00 per shift allowance
Highway Toll Gate: ₹1,850.00 fast-tag receipt

2. The 24-Hour Chronological Operational Walkthrough

06:00 AM • SHIFT MUSTER & COMPLIANCE GATE

Stage 1: Driver Roster & Statutory Verification

Driver **Rajesh Kumar** reports for the Morning Shift (06:00 – 14:00). Dispatcher opens `/shifts`. The system automatically performs a 3-way compliance validation before confirming the roster:

- **Licence Check:** Validates DL-042018009124 is active and not suspended.
- **Vehicle Check:** Validates KA-19-MG-8802 Insurance and Mining Pass are valid.
- **Double-Booking Lock:** Verifies neither truck nor driver is allocated to another pit.

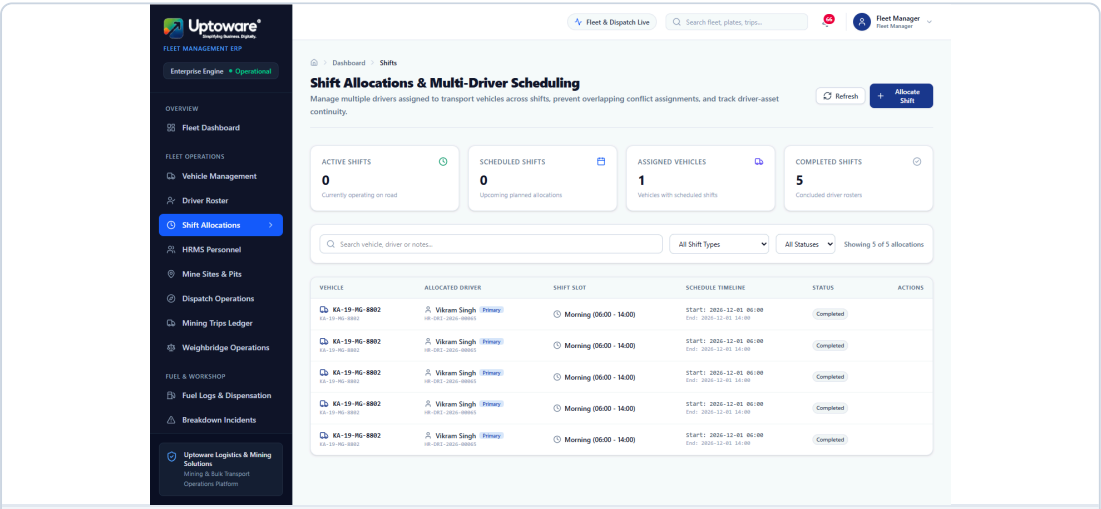


Figure S.1: Shift Allocation Board validating single-vehicle assignment.

06:15 AM • PIT LOADING & DISPATCH

Stage 2: Digital Trip Token Generation (TR-2026-00412)

Excavator EX-300 loads KA-19-MG-8802 with Iron Ore Lump at Kudremukh Bench 4. The pit supervisor issues digital trip **TR-2026-00412** in `/dispatch`. Status changes to '*In Transit*'. The system records the departure timestamp and GPS geofence departure.

07:10 AM • SIDING WEIGHBRIDGE SCALE

Stage 3: Scale Integration & Payload Netting (WB-2026-00388)

The tipper arrives at Railway Siding #2. The automated scale captures Gross Weight, driver unloads at the crusher hopper, and scale captures Tare Weight:

Scale Reading	Weight (kg)	Weight (MT)	System Verification
Gross Weight (Loaded)	50,400 kg	50.40 MT	Captured via digital scale terminal
Tare Weight (Empty)	18,000 kg	18.00 MT	Matches vehicle tare baseline (±1%)
Net Delivered Payload	**32,400 kg**	**32.40 MT**	**Within 35.0 MT safe limit (No Overload)**

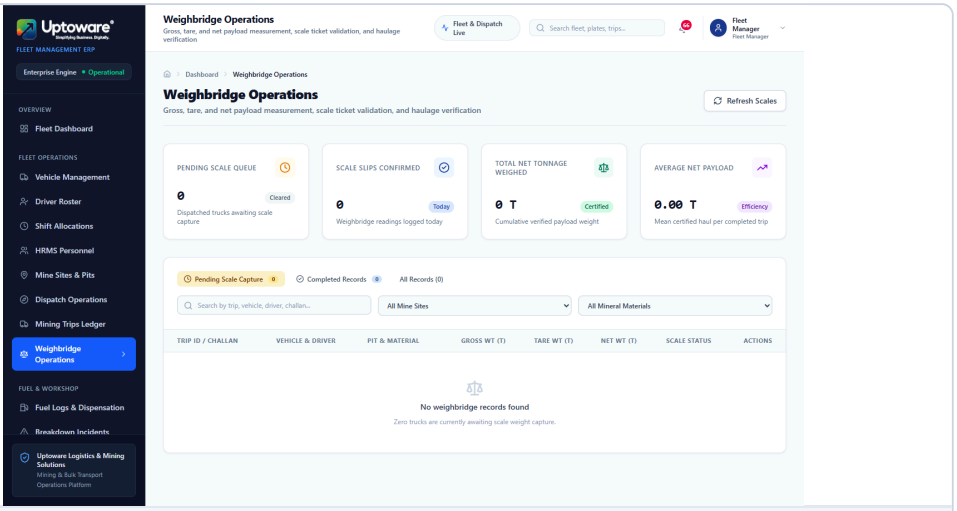


Figure S.2: Weighbridge Netting Console verifying 32.40 MT delivery.

08:30 AM • PIT BUNK FUELING

Stage 4: Bowser Dispense & Mileage Calculation (FL-2026-00192)

After completing its second cycle, the truck stops at Internal Pit Bowser #1 for refueling. The operator records fuel ticket in `/fuel`:

- **Litres Pumped:** 180.0 Litres High-Speed Diesel
- **Unit Rate:** ₹92.50 per Litre = **₹16,650.00 Total Cost**
- **Odometer Reading:** 142,380 km (Previous: 141,993 km → Distance: 387 km)
- **Calculated Efficiency:** 387 km / 180 L = **2.15 km/L** (Healthy efficiency)

11:45 AM • WORKSHOP INCIDENT & REPAIR

Stage 5: Unscheduled Breakdown & Maintenance Order (WO-2026-00109)

To demonstrate how Uptoware handles fleet disruptions, highlight companion truck **KA-19-MG-7001** which reports a hydraulic hose rupture in `/breakdowns`:

- **Severity:** Major • Location: Bench 3 South Ramp
- **Action:** Converted to Maintenance Work Order **WO-2026-00109**.
- **Parts Issued:** 2x Hydraulic Seal Kits issued from `Stores Central` (Inventory debited).
- **Cost Impact:** ₹4,200.00 parts + ₹1,500.00 labour charged directly to truck's ledger.

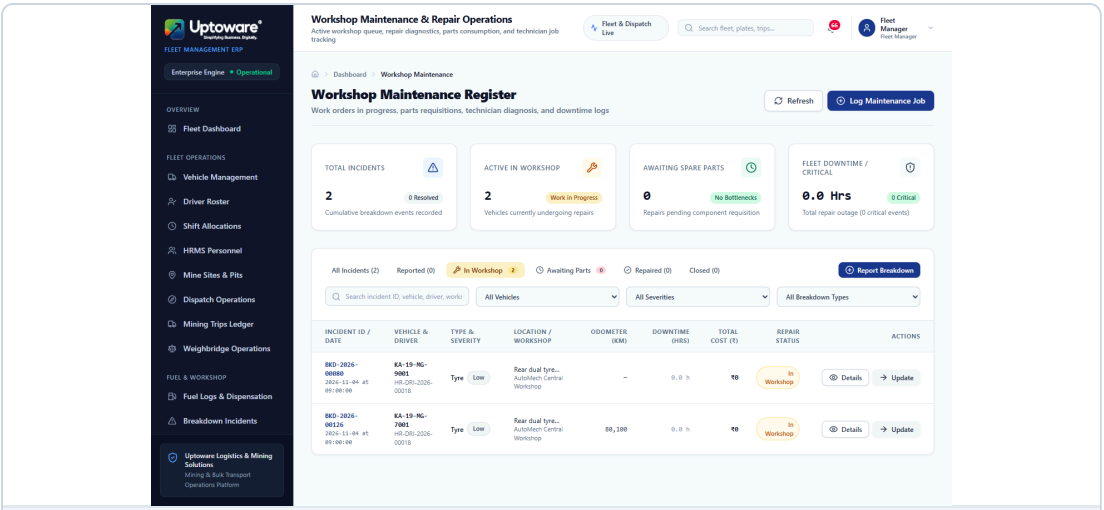


Figure S.3: Maintenance Job Card tracking parts issue and repair completion.

02:00 PM • FIELD ROAD EXPENSE & MD APPROVAL

Stage 6: Toll Expense (EXP-2026-00564) & Dual-Control Authorization

Driver Rajesh Kumar submits highway toll charges incurred during delivery to the railway siding:

- **Amount:** ₹1,850.00 (Toll + Municipal Mineral Transit Fee)
- **Workflow State:** Automatically routed to **Managing Director Approvals Inbox** in `/approvals` because amount exceeds ₹1,000 threshold.
- **Approval:** Managing Director reviews receipt photograph, clicks **Approve**, and system stamps approval with timestamp and user ID.

05:00 PM • AUTOMATED BILLING GENERATION

Stage 7: Sales Invoice Generation (SINV-2026-00088)

In `/revenue`, the accountant runs the daily billing cycle for Tata Steel Mining. The system aggregates all weighbridge slips logged today for Kudremukh North Pit:

- **Tonnage Hauled by KA-19-MG-8802:** 3 Completed Trips x 32.40 MT = **97.20 MT**
- **Contract Tariff:** ₹450.00 per Metric Ton
- **Total Revenue Generated:** 97.20 MT x ₹450 = **₹43,740.00**
- **Invoice Status:** Generated with GST breakdown and sent for client wire settlement.

08:00 PM • END-OF-DAY EXECUTIVE PROFITABILITY AUDIT

Stage 8: The Single-Truck Profit & Loss Statement

The managing director opens `/profitability` to inspect the true net operating profit generated by Tipper KA-19-MG-8802 across today's operations:

P&L Account Line Item	Reference Voucher	Debit / Cost	Credit / Revenue
Hauling Revenue (97.20 MT @ ₹450/MT)	SINV-2026-00088	—	₹43,740.00
Diesel Fuel Dispensed (180.0 Litres)	FL-2026-00192	₹16,650.00	—
Driver Shift Allowance & Bhatta	HR-ATT-2026-001	₹450.00	—
Highway Tolls & Transit Permit	EXP-2026-00564	₹1,850.00	—
Daily Amortized Maintenance Reserve	WO-ACC-2026-01	₹2,500.00	—
Tyre Wear & Depreciation Provision	CAP-DEP-2026-01	₹8,550.00	—

P&L Account Line Item	Reference Voucher	Debit / Cost	Credit / Revenue
NET OPERATING PROFIT (31.4% MARGIN)	**DAILY SUMMARY**	**₹30,000.00**	**₹13,740.00**

DEMONSTRATION CLIMAX SUMMARY TO SPEAK

"Gentlemen, look at this table. On a single tipper, you generated ₹43,740 in revenue, spent ₹30,000 in fuel, tolls, and maintenance, and banked ₹13,740 in net operating profit today. Multiply that by 50 trucks across 300 operating days, and you are looking at over ₹2 Crore in profit that is completely visible, audited, and protected against pilferage."